<u>Design and Implementation of a Mini AI Accelerator Using Verilog HDL on FPGA</u>

Chapter 1: Introduction

Introduction

Artificial Intelligence (AI) has become one of the most important technologies in modern computing systems. AI applications such as image recognition, speech processing, autonomous vehicles, recommendation systems, and natural language processing require a large number of arithmetic operations. Most of these computations are based on multiplication and accumulation processes performed within neural networks.

Traditional processors can execute AI algorithms, but specialized hardware accelerators provide higher speed and improved efficiency. AI accelerators are dedicated hardware architectures designed to perform neural network computations faster by executing multiple arithmetic operations in parallel.

The fundamental operation in most neural networks is the Multiply-Accumulate (MAC) operation. A MAC unit multiplies input values with corresponding weights and accumulates the results to produce neuron outputs. Modern AI processors and accelerators consist of thousands or millions of MAC units working together.

This project presents the design and implementation of a Mini AI Accelerator using Verilog HDL. The design begins with basic digital building blocks such as Half Adders and Full Adders, followed by higher-level modules including Ripple Carry Adders, Multipliers, MAC Units, and Artificial Neurons. These modules are integrated to form a Mini AI Accelerator capable of performing parallel neuron computations.

The complete design was developed and verified using AMD Vivado and targeted for the Artix-7 FPGA platform. Behavioural simulation and RTL analysis were used to validate the functionality of each module.

Chapter 2: System Architecture

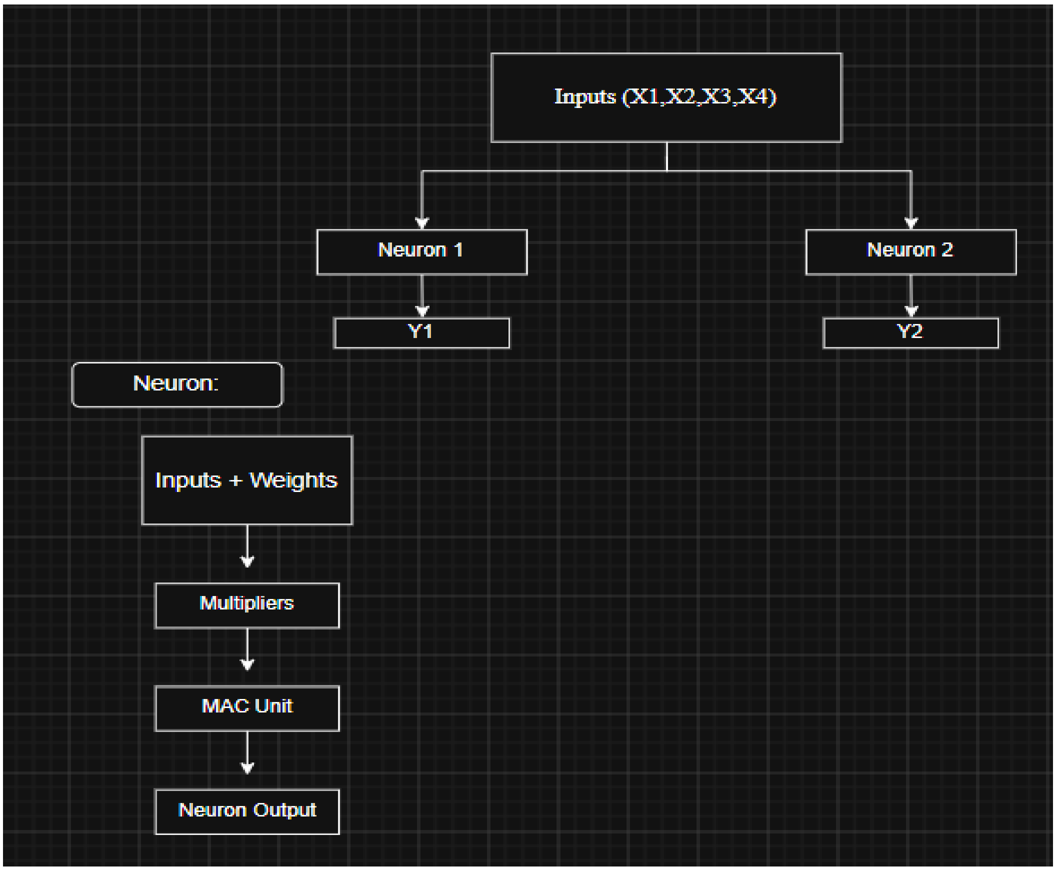

Figure 2.1: Architecture of the proposed Mini AI Accelerator. The accelerator consists of two neurons operating in parallel. Each neuron performs weighted computations using multipliers and a MAC unit to generate the final output.

Chapter 3: Module Description

Module	Function
Half Adder	Adds two single-bit numbers
Full Adder	Adds three single-bit numbers
Ripple Carry Adder	Performs multi-bit addition
Multiplier	Multiplies two 4-bit numbers
MAC Unit	Performs Multiply-Accumulate operation
Mini AI Accelerator	Executes two neurons in parallel

Chapter 4: Software and Hardware Requirements

Requirement	Details
Language	Verilog HDL
Simulator	Vivado Simulator
Design Tool	AMD Vivado 2025.2
FPGA Target	Nexys A7 (Artix-7)
Operating System	Windows 11

Chapter 5: Verilog Implementation

Verilog Implementation

The proposed Mini AI Accelerator was implemented using Verilog HDL following a hierarchical design methodology. The implementation started with basic arithmetic circuits and gradually progressed to higher-level AI computation modules.

1. Half Adder
   - Performs addition of two single-bit inputs.
   - Generates Sum and Carry outputs.
2. Full Adder
   - Extends Half Adder functionality by including Carry-In input.
   - Produces Sum and Carry-Out outputs.
3. Ripple Carry Adder
   - Constructed using multiple Full Adders.
   - Performs multi-bit binary addition.
4. 4-Bit Multiplier
   - Multiplies two 4-bit binary inputs.
   - Produces an 8-bit product output.
5. MAC (Multiply-Accumulate) Unit
   - Performs multiplication followed by accumulation.
   - Fundamental building block used in AI hardware accelerators.
6. Neuron
   - Accepts inputs and corresponding weights.
   - Performs weighted summation using MAC operations.
   - Produces neuron activation output.
7. Mini AI Accelerator
   - Integrates two neurons operating in parallel.
   - Demonstrates the concept of parallel neural computation.

The hierarchical approach improved modularity, reusability, and ease of verification during simulation.

Chapter 6: Simulation Result

Simulation Results

Behavioral simulations were performed using AMD Vivado Simulator to verify the correctness of each module.

The following simulations were successfully verified:
• Ripple Carry Adder
• 4-Bit Multiplier
• MAC Unit
• Neuron
• Mini AI Accelerator

RTL schematics were generated for major modules to verify the synthesized hardware structure.

The obtained waveforms matched the expected theoretical outputs, confirming the correctness of the Verilog implementation.

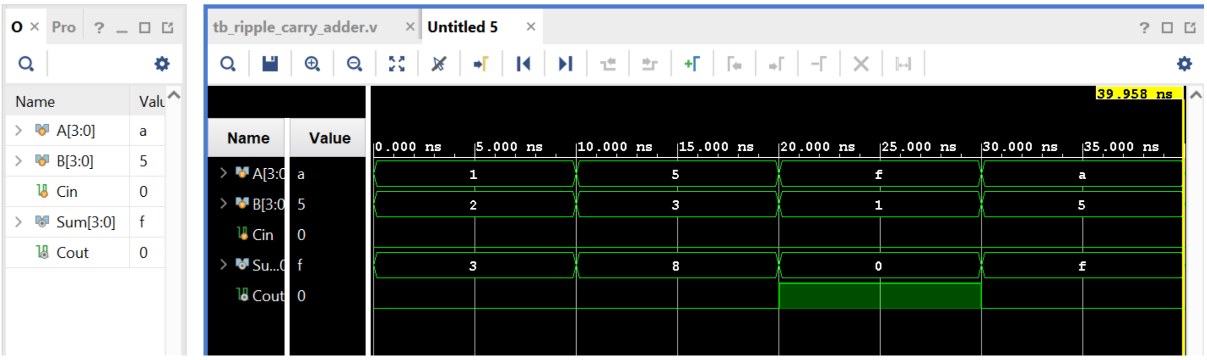

Figure 6.1: Ripple Carry Adder Simulation Waveform

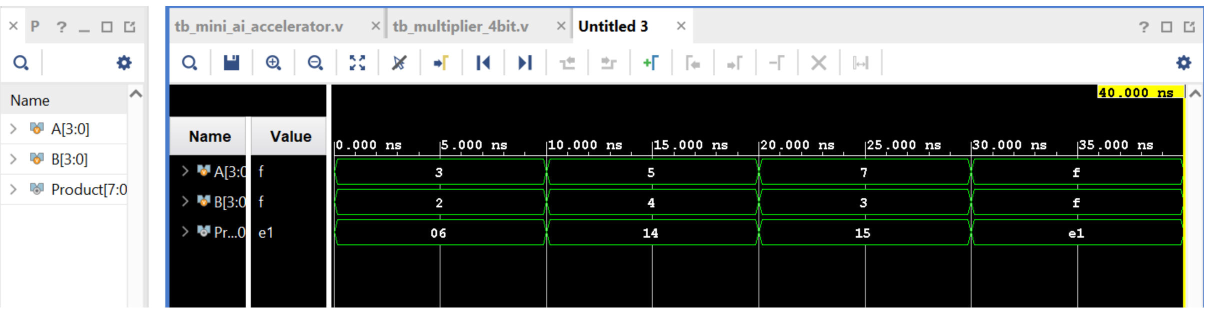

Figure 6.2: Multiplier Simulation Waveform

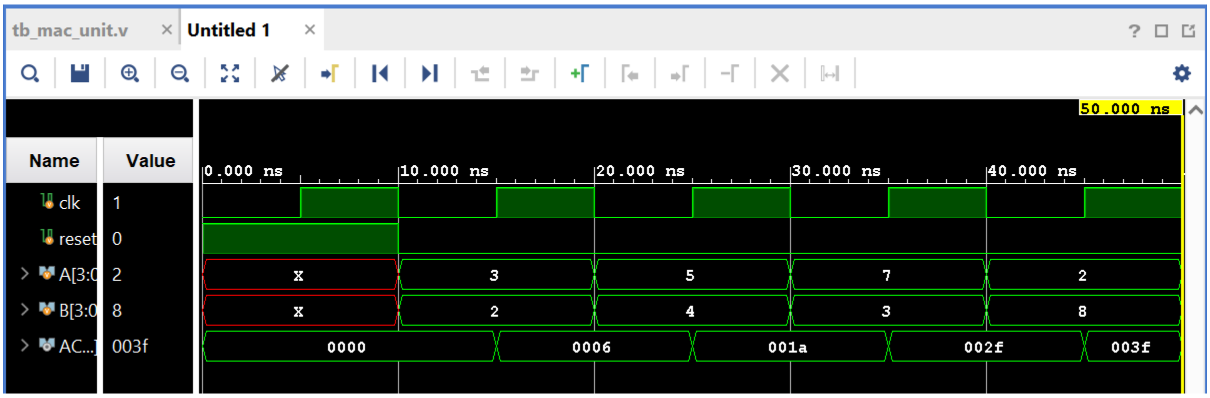

Figure 6.3: MAC Unit Simulation Waveform

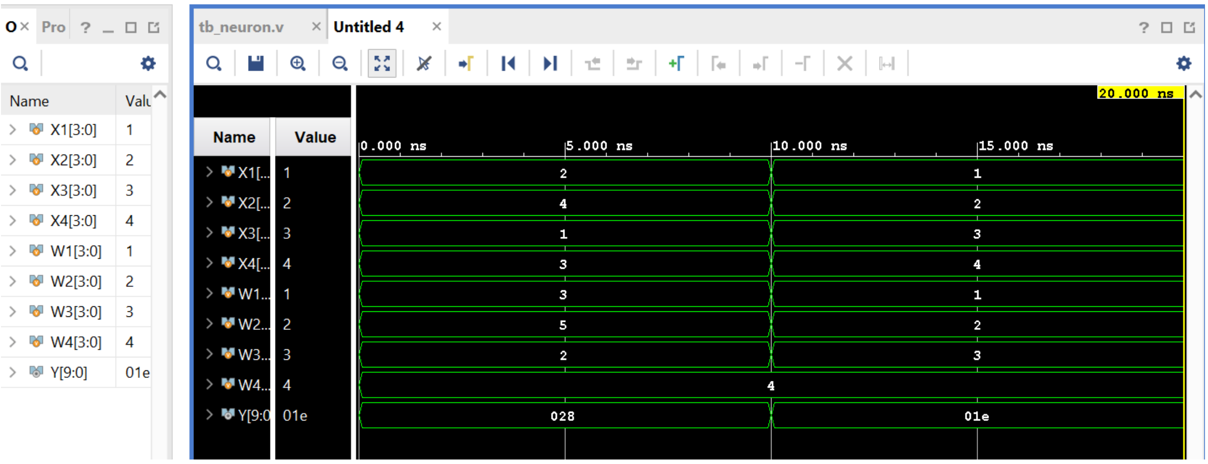

Figure 6.4: Neuron Simulation Waveform

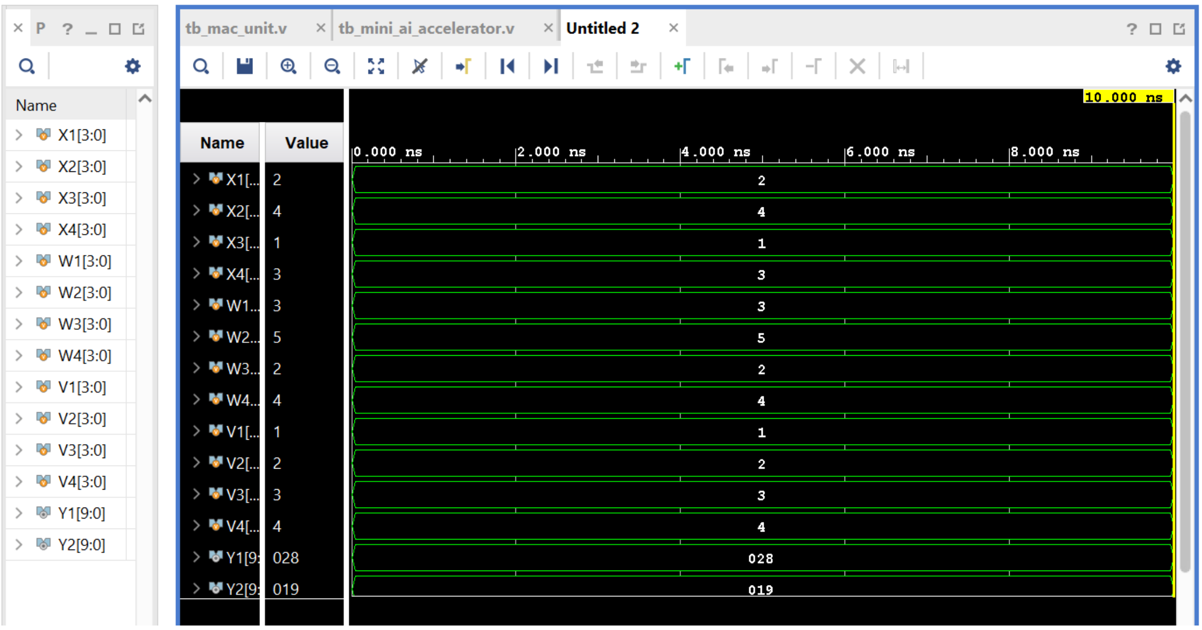

Figure 6.5: Mini AI Accelerator Simulation Waveform

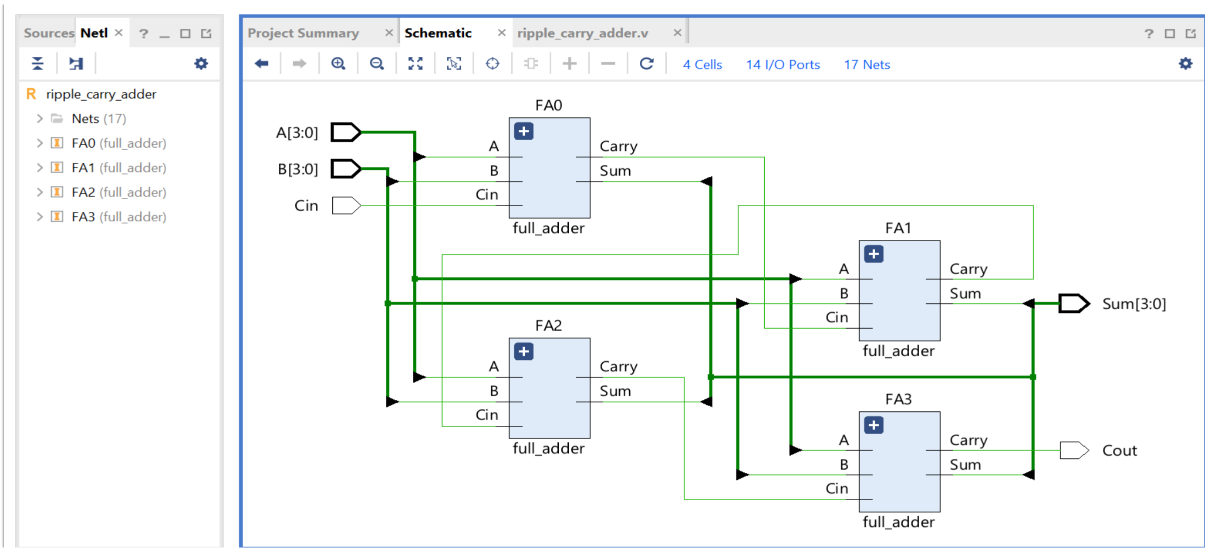

Figure 6.6: RTL Schematic of Ripple Carry Adder

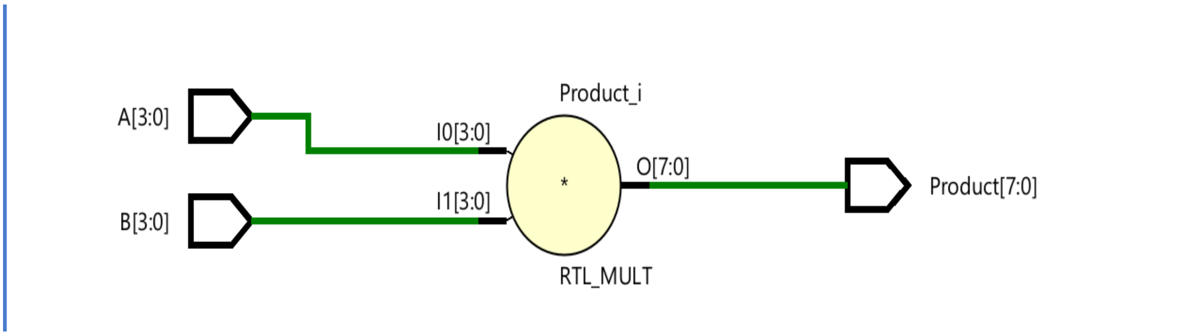

Figure 6.7: RTL Schematic of Multiplier

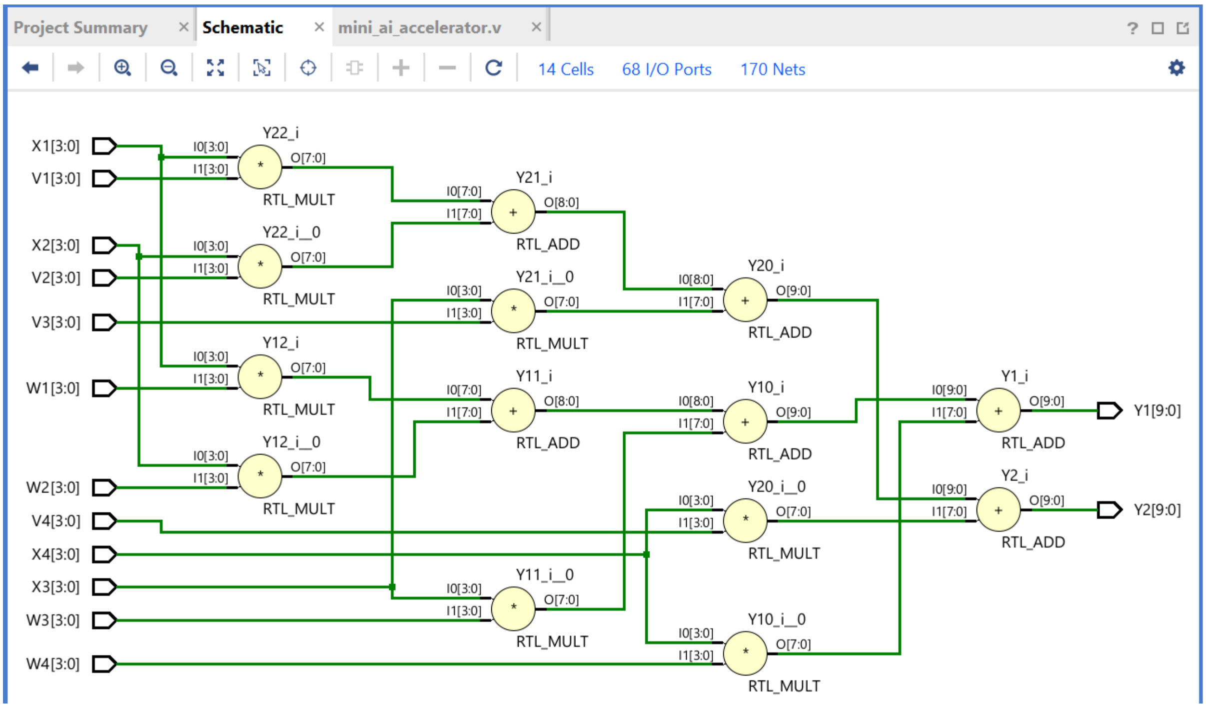

Figure 6.8: RTL Schematic of Mini AI Accelerator

Chapter 7: Applications

1. Edge AI Systems
    - Used in low-power AI processing devices.
2. FPGA-Based AI Prototyping
    - Useful for testing neural network architectures before ASIC implementation.
3. Embedded Systems
    - Can be integrated into IoT and smart devices.
4. Signal Processing
    - Supports arithmetic-intensive computations.
5. Educational and Research Purposes
    - Demonstrates the basic architecture of AI hardware accelerators.

Chapter 8: Future Scope

Future Scope

1. Increase the number of neurons to support larger neural networks.
2. Implement matrix multiplication engines.
3. Introduce pipelining to improve performance.
4. Deploy the design on a Nexys A7 FPGA board.
5. Perform timing, area, and power analysis.
6. Extend the design towards ASIC implementation.
7. Develop support for convolutional neural network operations.

Chapter 9: Conclusion

This project successfully implemented a Mini AI Accelerator using Verilog HDL. The design was developed hierarchically using Half Adders, Full Adders, Ripple Carry Adders, Multipliers, MAC Units, and Artificial Neurons. Behavioural simulations and RTL analysis verified the correctness of the design.

The project demonstrates the fundamental principles used in modern AI hardware accelerators and provides a strong foundation for future FPGA and ASIC-based AI hardware development. The implementation enhanced understanding of digital design, hardware description languages, and AI accelerator architecture.

Chapter 10: References

[1] AMD Vivado Design Suite User Guide.

[2] M. Morris Mano, Digital Design.

[3] Samir Palnitkar, Verilog HDL.

[4] FPGA Prototyping by Verilog Examples.

[5] Research articles on AI Accelerators and MAC Architectures.